



Principal Component Analysis (PCA) and PCA vs FA Transcript

Hello everyone, welcome to lesson 1 video 3. In this video, we will compute principal component analysis or what we call as PCA. Although both factor analysis and PCA are data reduction methods, they are based on different ideas. In this video, we are going to learn about PCA vs factor analysis, though both of them are very similar tools.

They work on the same concept of dimensional reduction and they actually reduce dimension. You have already looked into factor analysis. If you have common or commonalities between input variables, you can actually combine them together and create factors.

Similarly, in PCA, which is also called as principal component analysis, we combine all the information in higher dimensional space, project that data into lower dimension thinking that in lower dimension, we project exactly in a dimension such that we have maximum variance in the data but also we have certain dimension which captures those maximum variances. So that's the underlying principle between PCA but you will have that understanding more in this entire video. So PCA basically stands for principal component analysis.

Now this is again a data reduction technique. It transforms the input data and minimises the number of variables that it has. So what it does is, if you have a lot of correlated data just like what you have here, you have a lot of correlated data, it projects all the data onto a dimension such that the variance of the projected data maximises.

So it chooses an axis on which after projecting the data, the variance of that projected data is maximised. So it doesn't choose any arbitrary axis. It wants the variance of the data to be maximised after projection.

So it cannot choose, let's say, PC2 because on PC2, the variance might be only from here to here or the range of the data, right? While if we project all of these blue points on the red points, if you think of this would be the first projection, this would be the second projection, here for this point you have a projection here. So the length of entire projection would be this, right? Whatever I'm highlighting. So on this axis, the variance of the projected data would be much more than what you will have on the PC2.

So the idea is that we choose certain principal components or certain new accesses such that after projecting the data onto those accesses, we maximise the variance. Now after let's say we have 100% of the variance in our two dimensional data, right now this entire data is two dimensional, right? Let's say for example, this could be, if you think of x and y axis, this could be age of individual and on y axis could be experience, okay? And age and experience are highly correlated. Now think of if you project this data on a new axis altogether, which is red, and these are the projections, you might get almost 90% of variance captured in your input data by red line, while 10% of the variance could be captured by PC2.



Now PC1 because it captures 90% of the variance or 95% of the variance or very high variance, why do we need PC2 in that case, we would be good enough with PC1, we actually do not need PC2. So this is the entire essence. Now because we started with two dimension, let's say age and experience on two accesses, now we, once we project the data on the new two accesses, if one of the accesses captures 90-95% of the variance, why do we need the second access on which we have projected the data, right? So in this case, we would be using PC1 as the only dimension, the projected data.

And once we have that, we will use that one dimension instead of higher dimension data or let's say, in this case, we had two dimension, we have projected the entire data on PC1 and PC2 and we got that 90% of the variance is on PC1. Hence we are using only PC1, we will not care about PC2. So this way, the entire data gets reduced or the dimension of the data gets reduced.

Okay, I hope this is clear. Now one other thing which is not written on this particular slide is these PC1, PC2 that we find out using PCA, they are orthogonal to each other. It doesn't look like they are orthogonal, but because I wanted to explain you the concept of projecting the data onto new accesses, but the new accesses that it forms, if your data has two dimensions, let's say age and experience, on y-axis it has experience, on x-axis it has age and this is all the data that you have.

So it will project, PCA will project the data onto two new accesses. So these two new accesses are PC1 and PC2. Now these two new accesses, if you have a situation like the data is highly correlated, you actually would get most of that variance explained by PC1.

And these two PCs or principal components, they are always orthogonal to each other. That's what you will look into and one of the HTMLs that we'll be providing for you and during the reading material or as I'm reading material, so you go through it, interact on that and learn about it. Now, there are critical differences between PCA and what we have as factor analysis.

PCA explains the total variance in our data. So let's say what proportion of total variance is being explained by our principal components. So that's what PCA does.

But factor analysis explains the common variance. So let's say we have too many features, what is the common between them? It explains the commonalities between the variables. It doesn't try to explain the total variance, but it tries to explain the total common variance between them.

Okay, let's say there are 10 features, what is the common variance between those 10 features? So feature or factor analysis is all about commonalities, while PCA is all about decomposing total variance into new features that we find out as PC1, PC2. Now, this is also another way of visualising it. The thing about PCA, PCA captures the



entire full variance of, let's say we have two dimensions V1, V2 or two variables V1, V2 and you have the purple one as V1, this orange one as V2.

So PCA tends to describe the entire variance, which is the outer green circle, if you think While factor analysis, you have V1 and V2, factor analysis tries to understand what is the common between them. What is the common variance between them, which could be this green piece of thing that you are seeing. This is all about differences between PCA and PCA and FA or principal component analysis and factor analysis.

I hope this entire difference is, you are able to understand it really well. But if you have to keep a quick check on what to use when, PCAs are used when you have the primary problem is to reduce the data. If you have too many features or too many variables and you are working on a data when you have hundreds of variables, let's say you are trying to find out the credit score of an individual and you have age, financial status, bank statements, XYZ, that you have too many data points, right? So your idea is to reduce the data in that cases.

So PCA is well suited in those cases. Second, if you would want to create uncorrelated features from the input data, then also PCA is well suited in those cases as well. If you are looking for commonalities in the data, then factor analysis is well suited for.

If you are looking for unobservable underlying latent variables within the data, or let's say for example, what could define the input data? What is the common subject that the data has, then you should use factor analysis. And they, as I told you, they work on finding common things between variables. So if you have a bunch of variables, they would find which one are talking about the same common subject.

So that's the basic difference between PCA and FA. I hope you are able to understand what exactly is PCA, what exactly is FA in this entire module. In summary, PCA and factor analysis reduces dimensions, but they are used for serving different purposes.

In the next video, we will learn how to interpret our output of factor analysis

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